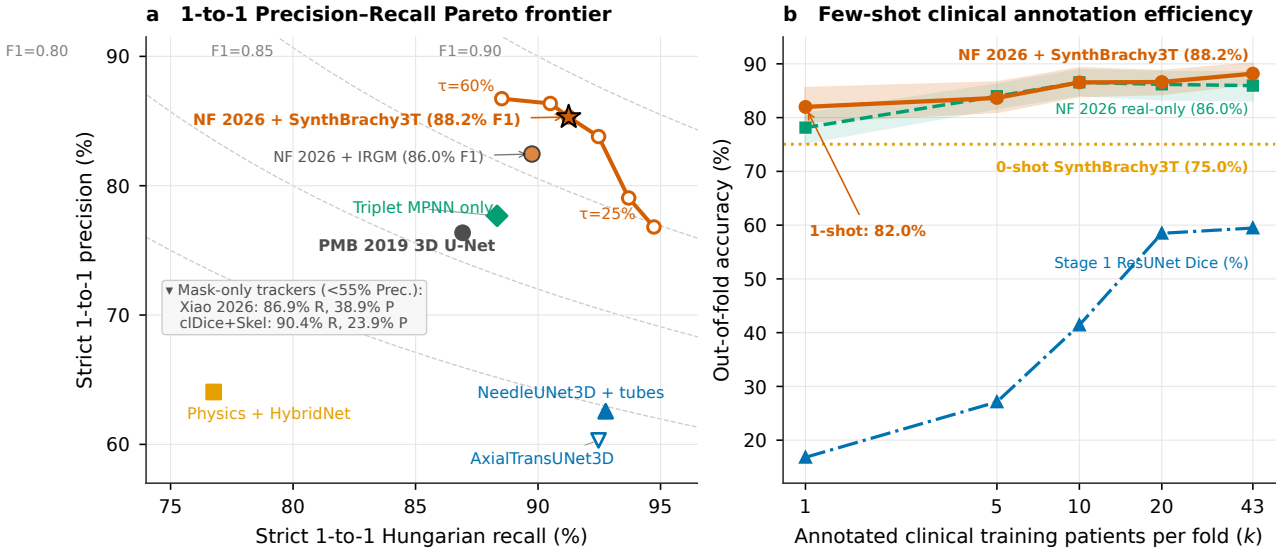
a 1-to-1 Precision–Recall Pareto frontier
F1=0.80
F1=0.85
F1=0.90
τ=60%
NF 2026 + SynthBrachy3T (88.2% F1)
NF 2026 + IRGM (86.0% F1)
Triplet MPNN only
τ=25%
PMB 2019 3D U-Net
▾ Mask-only trackers (<55% Prec.):
Xiao 2026: 86.9% R, 38.9% P
clDice+Skel: 90.4% R, 23.9% P
Physics + HybridNet
NeedleUNet3D + tubes
AxialTransUNet3D
Strict 1-to-1 precision (%)
Strict 1-to-1 Hungarian recall (%)
b Few-shot clinical annotation efficiency
NF 2026 + SynthBrachy3T (88.2%)
NF 2026 real-only (86.0%)
0-shot SynthBrachy3T (75.0%)
1-shot: 82.0%
Stage 1 ResUNet Dice (%)
Out-of-fold accuracy (%)
Annotated clinical training patients per fold (k)